Software Engineering & Project Management (SEPM)

Energy & Utility Management System (EUMS)

Billing, metering and customer service for electricity, water and gas

Test Cases and Results

Software process model: Waterfall · Combined SEPM + DBMS course project

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Date	October 2026

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1. Testing Performed

Testing so far consists of **unit tests** for the billing calculation and **functional (integration) tests** that drive the real Flask application through its pages with a fresh, seeded SQLite database for each test. All tests are automated with pytest and can be re-run with `python -m pytest -v` in `04_Source_Code`. The actual results below were produced by running the suite: **20 passed in 12.84s**. Manual exploratory testing of every page was also done for each role (screenshots in Appendix B).

1.1 Test-case table

ID	Feature	Input / steps	Expected result	Actual result	Status
UT-01	Slab billing	250 kWh, residential electricity (0-100 @3.50, 100-300 @5.00), fixed 50, tax 5%	Energy 1,100.00; tax 57.50; total 1,207.50	As expected	**PASS**
UT-02	Slab billing	0 units	Only fixed + tax = 52.50, no slab rows	As expected	**PASS**
UT-03	Slab billing	Exactly 100 units (slab edge)	Only first slab used; total 420.00	As expected	**PASS**
UT-04	Slab billing	400 units (reaches 300+ slab)	Energy 2,050.00; total 2,205.00; 100 units in top slab	As expected	**PASS**
UT-05	Slab billing	-5 units	ValueError raised	As expected	**PASS**
UT-06	Slab billing	15.5 kL water, no tax	Energy 146.00; total 186.00	As expected	**PASS**
IT-01	Login	admin / admin123	Redirect to /staff/dashboard; dashboard loads	As expected	**PASS**
IT-02	Login	admin / wrong	Stays on login with 'Invalid username or password'	As expected	**PASS**
IT-03	Access control	Open /staff/bills without logging in	Redirect to /login	As expected	**PASS**
IT-04	Access control (RBAC)	Customer opens staff page; employee opens admin-only Tariffs	HTTP 403 for both; employee can open Add Reading	As expected	**PASS**
IT-05	Data isolation	Sushrut (C1001) opens a bill that belongs to Sumit (C1002)	HTTP 403 on bill and pay page	As expected	**PASS**
IT-06	END-TO-END: reading → bill -> payment	Employee adds +250 kWh on Sushrut's meter; Sushrut pays 1,207.50 by UPI	Bill of 1,207.50 created (unpaid); after payment status = paid and 'Payment received' notification	As expected	**PASS**
IT-07	Meter reading validation	Current reading 10 below previous	Trigger error 'Reading cannot be lower...'; no bill created	As expected	**PASS**
IT-08	Meter reading validation	Reading date 2020-01-01	Trigger error 'Reading date must be after...'	As expected	**PASS**
IT-09	Payment validation	Sushrut pays outstanding + 100	Trigger error 'Payment exceeds outstanding balance'; bill not paid	As expected	**PASS**
IT-10	Counter payment	Billing officer pays half the bill in cash	Bill status = partial	As expected	**PASS**

IT-11	Outage alerts	Admin reports electricity outage in Alambagh	Every Alambagh electricity customer notified; Hazratganj customers not notified	As expected	**PASS**
IT-12	Complaints	Sushrut raises a complaint; technician marks it resolved	Complaint visible to staff; status resolved with timestamp	As expected	**PASS**
IT-13	Customer registration	Register Sushrut Lad again with the same email	Error 'already exists'; customer count unchanged	As expected	**PASS**
IT-14	Tariff management	Admin changes slab rate to 9.99	New rate saved; old bill total unchanged	As expected	**PASS**

Table 1: Test cases with expected and actual results (UT = unit test, IT = integration/functional test)

1.2 Evidence

The full pytest log is in `05_Testing/Test_Evidence/pytest_output.txt`. The screenshots below show the same rules being enforced in the user interface.

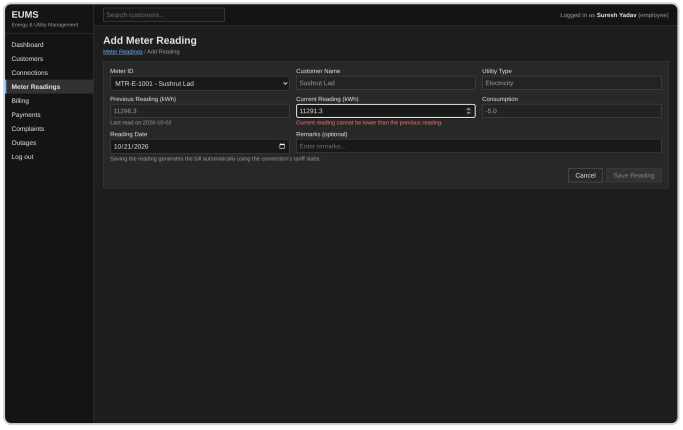


Figure 1: IT-07: lower reading blocked

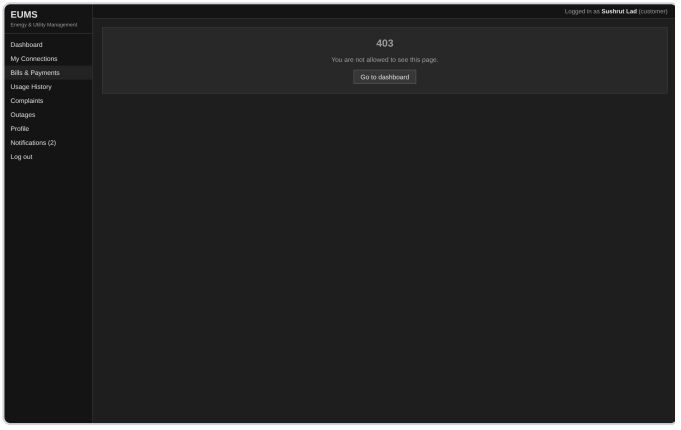


Figure 2: IT-04: customer opening a staff page gets 403

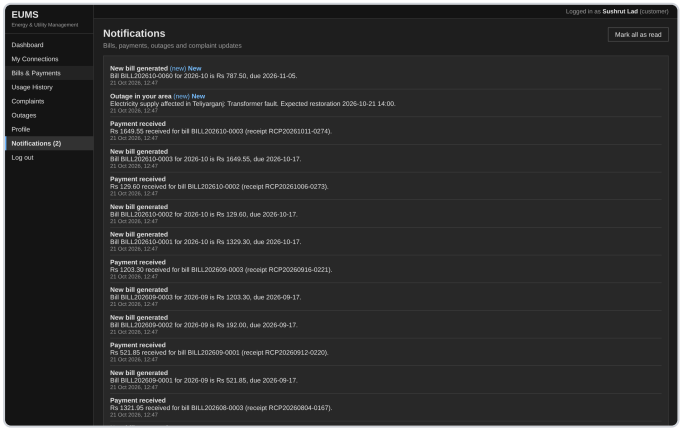


Figure 3: Trigger-generated notifications for the customer

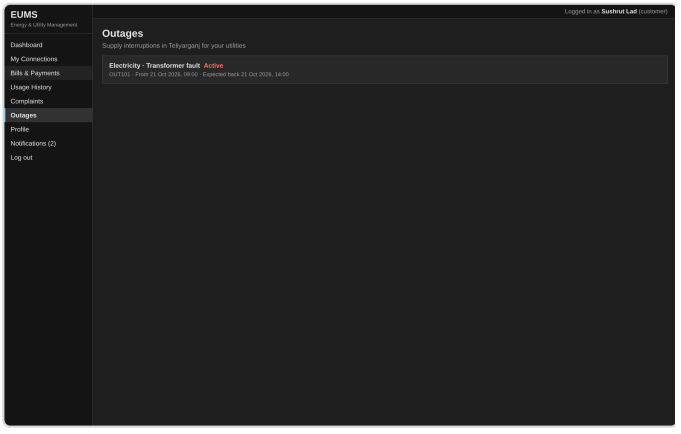


Figure 4: IT-11: outage shown to customers in that area

1.3 Testing still to be done

- System testing of the complete workflow by a person outside the team, with a written checklist.
- Boundary tests for forms (very long text, special characters) and a basic security test (SQL injection attempts; all queries are already parameterised).
- Performance check with 1,000+ customers of generated data (NFR-07).
- User-acceptance testing with 3–5 classmates acting as customers and employees.